Homework Assignment 3

The third homework assignment is due on **October 27 at 11:59PM**. Please closely follow instructions listed in the syllabus.

Note, for this homework assignment you are expected to use **tidyverse functions only**.

You will be working with the datasets containing demographic information of midwest counties from 2000 US census. **Dataset_1**, **Dataset_2**, and **Dataset_3** are available on Courseworks.

Dataset_1 contains the following variables:

- **PID** - Unique county identifier
- **county** - County name
- **area** - Area of county (units unknown). The values of this variable are spread across 5 columns (IL - WI)
- **state** - State to which county belongs to. The column names (IL - WI) represent the values of this variable
- **poptotal** - Total population of the county

Dataset_2 contains the following variables:

- **county** - County name
- **state** - State to which county belongs to
- **race** - Total population for each racial group (White, Black, American Indians, Asians, Other races)
- **inmetro** - A factor with levels 0 and 1 to indicate whether county considered in a metro area
- **category** - Miscellaneous

Dataset_3 contains the following variables:

- **county** - County name
- **state** - State to which county belongs to
- **popadults** - Number of adults
- **perchsd** - Percent of population with high school diploma
- **percollege** - Percent of college educated population
- **percprof** - Percent of population with professional degree
- **percbelowpoverty** - Percent of population below poverty line
- **percchildbelowpovert** - Percent of children below poverty line
- **percadultpoverty** - Percent of adults below poverty line
- **percelderlypoverty** - Percent of elderly below poverty line

Note: The original description of the data was not documented and the current descriptions (provided by the `tidyverse` package) are based on speculation.

Note: Use `print()` function to display output results after each part of homework problems. For example, `print(vec1)`.

Problem 1 [42pt]

1. In **Dataset_1** the `area` variable is spread across multiple columns: the column names represent values of the `state` variable and the column values represent the values of the `area` variable. Transform/Tidy the dataset as follows: move the column names (IL-WI) to the `state` variable and the column values to the `area` variable. Then, remove all the missing value from the dataset. Name the new dataset `data1`.

2. In **Dataset_2** the `race` variable combines information about the total population for each racial group in following order: White, Black, American Indians, Asians, Other. Pull apart this column into multiple columns, by splitting wherever a separator character appears. Name these new variables `popwhite`, `popblack`, `popamerindian`, `popasian`, and `popother`, respectively. Name the new dataset `data2`.

3. Merge `data1` and `data2`: they should be joined by the `state` variable and the output result should contain all the observation from `data1`, regardless of whether they match or not. What are the dimensions of the output? How many missing values does it contain?

   Repeat the above procedure again by joining the datasets by the `state` and `county` variables.

4. Merge `data1` and `data2`: they should be joined by the `state` variable and the output result should contain all the observation from `data2`, regardless of whether they match or not. What are the dimensions of the output? How many missing values does it contain?

   Repeat the above procedure again by joining the datasets by the `state` and `county` variables.

5. Merge `data1` and `data2`: they should be joined by the `state` variable and the output result should contain all the observation from both `data1` and `data2`, regardless of whether they match or not. What are the dimensions of the output? How many missing values does it contain?

   Repeat the above procedure again by joining the datasets by the `state` and `county` variables.

6. Join `data1` and `data2` by the `state` and `county` variables such that the unmatched rows in either input dataset are not included in the result. Name the output dataset `data`.

7. Remove `PID` and `category` from `data` by selecting all variables except these two. Then, remove all duplicates from the dataset.

8. Add a new variable to the dataset (name it `popdensity`), which represents a population density in each county (in order to compute a population density you need to divide the total population by the corresponding area).

   Then, for each racial group, add a new variable to the dataset that will represent the percentage of the total population that belongs to this group (name them `percwhite`, `percblack`, `percamerindan`, `percasian`, and `percother`, respectively).

9. Join `data` and `Dataset_3` by the `state` and `county` variables such that the unmatched rows in either input dataset are not included in the result.

10. Add new variables to the dataset that correspond to the `percbelowpoverty` and `percadultpoverty` variables and represent the actual number of people that belong to these groups. Name these variables `popbelowpoverty`, `popadultpoverty`, respectively.

11. Move the `inmetro` variable to the end of the dataset; rearrange the `poptotal` variable so that it comes after the `area` variables; shift the `popdensity` variable to follow the `poptotal` variables.

12. What is the biggest/smallest state by the total population? What is the biggest/smallest state by the total area? What is the average population density in each state? In each state, compare the population densities of counties that are in metro area to counties that are not. Hint: you might want to use the grouping functions.

13. In all states combined, what is the average percentage of each racial group? In each state separately, what is the average percentage of each racial group? In each state separately, compare the average percentage of each racial group in metro areas to non-metro areas. Discuss the results you obtained.

14. For each state, calculate the average percentage of the total population who are below poverty. Now using the same measurement, compare the metro areas to non-metro areas in each state. What do you observe? Now filter out counties with less than 50000 population and repeat the previous step. What do you observe?

15. Plot a histogram of the `poptotal` variable: add labels (title, subtitle, x and y labels); adjust the x axis by zooming in on the interval [0, 100000] and changing the angle of ticks to 60; change the theme to minimal; change the colors of bins and bin outlines (it is up to you to choose colors); add a vertical line that represents the mean value of the variable.

16. Plot a boxplot of the `area` variable reflecting the categories of `inmetro`; add labels (title, subtitle, x and y labels); change the colors of boxplots; add black dots to these boxplots that represent the average area in these groups; make boxplots transparent (use transparency rate of 0.45).